



Ansh Kumar

☎ +91 91982 53088 ✉ Email 🌐 GitHub 🔗 LinkedIn 📄 LeetCode 🌐 Portfolio

Summary

Computer Science undergraduate with experience in full-stack development, machine learning, and software engineering. Skilled in building scalable web applications using React, TypeScript, Python, FastAPI, PostgreSQL, and modern development tools. Strong foundation in Data Structures & Algorithms, object-oriented programming, and system design with hands-on experience and contributing to real-world software projects

Education

Jaypee Institute of Information Technology (JIIT), Noida — B.Tech in CSE **2023 – Present**
CGPA: 7.0 / 10

St. Dominic Savio School, Lucknow, Uttar Pradesh — Class XII (ICSE) **2022**
Percentage: 88%

Technical Skills

- **Programming Languages:** C++, Python, JavaScript, TypeScript, SQL
- **Frontend:** React, Next.js, HTML, CSS, Tailwind CSS
- **Backend:** FastAPI, Node.js, REST APIs
- **Databases:** PostgreSQL, MongoDB, MySQL
- **Core CS:** DSA, OOP, OS, DBMS, Computer Networks
- **AI/ML:** Scikit-learn, Pandas, NumPy
- **Tools:** Git, GitHub, Linux, Bash, VS Code, Figma, Streamlit

Experience

High Performance Computing (HPC) Summer Intern - JIIT, Noida **June 2026 – July 2026**

- Developed a high-performance medical image segmentation pipeline using PyTorch on the Ramanujan Universe HPC cluster for brain tumor analysis.
- Accelerated image preprocessing and deep learning inference using OpenMP and CUDA, leveraging parallel computing on multi-core CPUs and NVIDIA GPUs.
- Executed large-scale training and inference jobs using PBS Professional, optimizing resource utilization and computational performance.

Projects

FraudLens — Credit Card Fraud Detection System ([Demo](#) | [GitHub](#)) *Python, Pandas, Scikit-learn, Streamlit*

- Built a fraud detection system on 1.29M+ transactions, engineering 18 features and handling class imbalance with SMOTE.
- Trained Logistic Regression, Decision Tree, Random Forest, and XGBoost models, achieving 99.43% ROC-AUC and 99% recall.
- Developed a Streamlit dashboard for real-time and batch fraud analysis with role-based access control.

SmartGym AI — AI-Powered Fitness Platform ([Demo](#) | [GitHub](#)) *React, TypeScript, PostgreSQL, MediaPipe*

- Built a full-stack AI fitness platform with 139 APIs, 33 PostgreSQL entities, and 92 React components for workout tracking and exercise analytics.
- Developed 9 MediaPipe-based exercise analyzers using 110K+ pose samples, achieving up to 99.58% accuracy.
- Implemented real-time exercise analysis with WebSockets, Python pose processing, JWT authentication, and role-based access control.

Certifications & Achievements

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional.
- Solved 200+ DSA problems across LeetCode and competitive programming platforms.
- Achieved a 1254 rating on CodeChef. ([CodeChef profile](#)).